

Project Documentation: Simple Payroll Calculator

1. Project Overview

The Simple Payroll Calculator is a comprehensive management system designed to automate the calculation of employee salaries, taxes, bonuses, and deductions. It features a modern web interface for HR/Admins to manage staff and a portal for employees to view their pay slips.

Key Features:

- Employee Management: CRUD operations for employee records.
- Dynamic Salary Calculation: Automated calculation based on base pay, overtime, and bonuses.
- Tax & Deduction Logic: Integrated tax bracket calculations.
- Pay Slip Generation: Exportable PDF and CSV reports.
- Admin Dashboard: Analytics on total company expenditure.

2. Technical Architecture

Frontend: HTML5, JavaScript (ES6+), and Vanilla CSS with Glassmorphism design.

Backend: Python Flask with modular routes (Blueprints).

Database: Supabase (PostgreSQL) for data persistence.

3. Database Schema

Table: employees

Column	Type	Description
id	UUID	Primary Key
emp_id	VARCHAR	Corporate ID
name	VARCHAR	Full Name
base_salary	DECIMAL	Monthly pay

4. Security & Best Practices

- Password Hashing: Use bcrypt for all credentials.
- Input Validation: Strict server-side validation for salary inputs.
- API Security: CORS restriction and session management.